

Adapting Before Contact: Compositional Contextual Inference for Safe Control with Frozen World Models

Research Proposal — v3

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Abstract

Robots that plan with learned world models make mistakes when the physics changes, for example when the table surface becomes more slippery or the object becomes heavier. Current methods fix this by updating the model from its prediction errors. They can only do so after the robot has touched the object, and they discard what they learned when the next episode starts. For tasks that are decided at the moment of first contact, such as striking a puck toward a target or lifting an unknown load, these methods therefore give no benefit at all.

We propose a different approach, based on a model of how humans learn motor skills [35]. The robot keeps a growing set of small correction models, one for each physical condition it has encountered (“memories”). It continuously estimates which condition it is currently in. It also learns, without labels, which visual features predict each condition, for example that a grey felt mat means high friction. When a known condition returns, the robot can select the right memory from what it sees, before it touches anything.

The method, CIRA (Context-Indexed Residual Adapters), adds these memories to any frozen world model. Memories for different factors, such as surface and load, can be combined, so a new combination of known factors can be predicted without new data. The same estimate of the current condition also drives a safety filter whose safety margins are tuned separately for each memory. The proposal contains two theoretical results and one measurement, each tested on simulated and real robot tasks:

- the error at first contact grows with how unpredictable the sequence of conditions is, while for existing methods it grows with the number of episodes in which the condition differs from training;
- a long-run bound on safety violations that holds whether or not the visual cues are reliable;
- a method that splits the robot’s recovery after a change into “recalling an old memory” and “learning something new”, which we measure over a long real-robot deployment.

1 Motivation and contributions

World models and their adaptation. A latent world model encodes camera images into a compact vector (the *latent state*) and predicts how that vector changes when the robot acts. Planners use these predictions to choose actions. Such models now plan manipulation tasks without task-specific training [6, 30, 77, 94]. When the physics at deployment differs from the training data, the predictions become wrong. Two recent methods address this. AdaJEPA updates part of the model with a few gradient steps on its recent prediction errors while the robot runs [86]. Sandwich-Residuals keeps the model fixed and trains small correction modules around it [78]. Both reset to the original model at the start of every episode, and both list keeping what was learned across episodes as future work. Keeping the gradient updates running indefinitely is not a good alternative. On long streams of data such updates gradually drift and degrade, and simply resetting from time to time works better [38, 70].

Real deployments repeat themselves. A robot in a lab, warehouse or kitchen rarely meets a completely new condition. It meets the same few surfaces, containers and tool states repeatedly, and these usually look different from each other. Humans make use of this. Research on human motor learning shows that people

keep separate motor memories for different conditions. They recall a memory when a condition returns instead of relearning it (*savings*). Learning a new condition does not erase the old one. Visual cues can select the right memory before the movement starts [34–36].

Running example. A robot arm strikes a puck toward a target on a table. The table is covered with one of three mats: felt (high friction), rubber (medium) or acrylic (low). The puck is sometimes empty and sometimes weighted. The mats and weights are swapped by people during the day. If the robot assumes the wrong friction, the puck stops short or slides off the table edge. The strike is a single contact, so the robot has no chance to correct its action after seeing the error.

Main idea. We treat adaptation as the question “which of the conditions I know is active now, or is this a new one?” rather than “how should I change my model parameters?”. The robot answers this question with a probability distribution over its memories (the *context posterior*). It updates only the memory that is likely to be active. This matters most at the first contact. At that point error-driven methods have received no error signal. The outcome of single-contact tasks is decided there, and so is much of the risk of unsafe behaviour.

Contributions.

- C1. Single-contact tasks as a test.** We define a class of tasks in which methods that only learn from prediction errors cannot outperform the unadapted model (Obs. 1). Examples are striking an object toward a target, lifting an unknown load and pushing near a table edge. On these tasks, only information available before contact can help.
- C2. The method, Cira.** A set of small probabilistic correction models (memories) on top of a frozen world model, with closed-form updates. One probability distribution over memories decides when a new memory is created, which memory is used and updated, and when the robot has returned to normal conditions. A noise model that depends on the state prevents the short, noisy moments at contact from being mistaken for a change of condition. Memories are organised by factor (for example surface and load), each with its own visual cue. As a result, an unseen combination of known factors can be predicted before contact.
- C3. Theory.** Proposition 1 bounds the total error at first contact. For our method the bound depends on how predictable the sequence of conditions is given the visual cues. For reset-based methods it grows with every episode that differs from training. Proposition 2 predicts how many steps the robot needs to identify a known condition. We state these predictions before running the experiments.
- C4. A safety filter that uses the memories.** The filter checks each planned action against every memory that is still reasonably likely. Its safety margin is tuned online, separately for each memory. When a known condition returns, the robot therefore also recovers the margin that was appropriate for it. Proposition 3 bounds the long-run rate of safety violations for any sequence of conditions, including when visual cues are misleading.
- C5. A measurement and a long deployment.** We split the improvement after a change of condition into two parts (Def. 2): the part that comes from selecting an existing memory, and the part that comes from changing a memory’s parameters. Following the human motor learning literature, we call these *apparent* and *proper* learning. We report this split over a real-robot deployment of at least 1000 episodes, in which people swap surfaces and containers on a schedule the robot does not know. We also release a benchmark of long sequences of changing conditions, built around four classic motor-learning experiments.

Scope. We study changes in physics (friction, mass, load, motor strength) while the image encoder stays fixed. Appearance matters only as a clue about which condition is active. We do not try to correct for changes in appearance that confuse the encoder itself, which is what AdaJEPA and Sandwich-Residuals address. We also assume that a safety constraint expressed in the latent space matches the real constraint. Whether it does is a property of the encoder that we do not study here [57].

2 Related work

Contextual inference in human motor learning. The MOSAIC model proposed that the brain holds several paired prediction and control models and chooses between them using both prediction accuracy and signals available before a movement [34, 87]. The COIN model [35] turned this into a full probabilistic model. It has the following parts:

- a set of contexts that can grow without limit, which tend to persist over time;
- one simple memory per context;
- visual or other cues whose probability depends on the context.

COIN explains several well-known effects by changes in which memory is used, not by changes in learning speed. These are savings, interference between opposing tasks, and the spontaneous return of old behaviour. It also predicted two effects that the standard two-rate model [76] cannot produce, and experiments confirmed both. One is the return of an old memory triggered by a few reminder trials (*evoked recovery*). The other is that the amount learned in one trial depends on the cue. Related models of animal learning make the same distinction between updating a memory and creating a new one [26, 27, 69]. People also learn separate parts of a task independently and recombine them [23].

One result is a warning for our design. Many purely visual cues do not help people separate motor memories [39], while differences in the planned movement do [74]. We therefore do not assume that a cue is useful. The robot learns how informative each cue is, and we vary cue reliability in the experiments.

Adapting world models while they run. AdaJEPA [86] updates encoder and predictor layers with about one gradient step per planning cycle on the PushT, PushObj and PointMaze tasks, including changes in mass and damping. Its authors note that adaptation is limited by what the pretrained representation can express. Sandwich-Residuals [78] reaches similar results while training only 1–3% as many parameters. More general test-time adaptation methods [80, 83, 85] degrade on long and repeating data streams [38, 70]. Contextual world models estimate a continuous context vector from recent experience [8, 47, 71]. Change-point-aware world models discard outdated training data after a detected change [90]. None of these methods stores separate conditions and recalls them later, and none acts before contact.

Probabilistic tools for switching dynamics. Our method combines existing tools:

- Bayesian last layers, which give fast closed-form updates of the final layer of a network [31, 33];
- methods that detect when the data-generating process changes [1, 32];
- mixtures of dynamics models whose number grows with the data, using a Chinese restaurant process (CRP) prior [41, 48, 64, 89];
- switching linear dynamical systems [21, 22, 52];
- the interacting multiple-model (IMM) filter from target tracking [12, 50, 61];
- lifelong learning with factored or modular models [20, 43, 62].

The closest prior method is MOLe [64], which keeps a CRP mixture of dynamics models for model-based reinforcement learning. It selects a model only from prediction errors, so it cannot act before contact. In online learning theory, “mixing past posteriors” [13] handles a small set of recurring experts. Our first-contact result builds on the same intuition.

Using vision to predict dynamics in advance. VcVBLL [14] adjusts a vehicle dynamics model using a hand-designed score for standing water in the camera image. Terrain-aware models use pretrained image features in a similar way [2, 15, 29]. Other methods train an image model online to predict physical properties measured by the robot’s own sensors [24, 54, 60]. Vision-language models can also provide prior estimates of physical properties [25, 84]. All of these map an image to a continuous physical quantity through a channel chosen in advance. CIRA instead maps the image to one of the robot’s own discovered conditions, and lets the observed dynamics overrule the image when they disagree.

	Latent visual world model	Keeps / recalls memories	Acts before contact	Cue learned during use	Combines factors	Safety uses memories	Calibrated under change
AdaJEPA [86], Sandwich-Res. [78]	✓	–	–	–	–	–	–
MOLe [64], IMGP [89]	–	✓	–	–	–	–	–
MOCA [32]	–	keeps only	–	–	–	–	partly
VcVBL [14], Phys2Real [84]	–	–	✓	– (fixed in advance)	–	–	✓
Uncertainty-aware latent filter [73]	✓	–	–	–	–	avoids unknown	✓
Latent world model + ACI filters [40, 67]	✓	–	–	–	–	one model	✓
COIN [35] (model of humans)	–	✓	✓	✓	–	–	–
CIRA (this proposal)	✓	✓	✓	✓	✓	✓	✓

Table 1: Comparison with the closest work, based on literature searches in September 2026. Each column is covered by some prior work. Our contribution is the combination, the single-contact task class and the first-contact result.

Safety in latent space. Several methods compute safety filters directly in the latent space of a world model, using reachability analysis or control barrier functions (CBFs) [5, 57, 65, 66, 88]. A CBF is a function h that is positive in safe states. A controller stays safe if it keeps h from decreasing too quickly. Seo et al. [73] treat states outside the training data as unsafe and steer away from them. AnySafe changes the safety constraint at runtime but not the dynamics [4]. Conformal prediction gives error bounds for learned predictors, and adaptive conformal inference (ACI) keeps these bounds valid when the data change over time [18, 28, 40, 51, 67, 79]. Adaptive and robust CBFs narrow down a single set of unknown parameters online [16, 53, 81, 82]. Multi-stage model predictive control is robust against a fixed set of candidate models [56]. What is missing is a filter whose set of candidate models is the set of remembered conditions that are currently likely. Such a filter can be cautious in new conditions, less cautious in recognised ones, and can store its calibration with each memory.

3 Problem setting

A frozen encoder maps an image o_t to a latent state $z_t = E(o_t) \in \mathbb{R}^d$. A frozen predictor gives the next latent state $\hat{z}_{t+1} = f(z_t, a_t)$ for action a_t . The robot runs one long sequence of episodes. It knows when an episode starts and ends, but it is never told which condition is active, and its memories are never reset.

In episode e an unknown context c_e^* is active. A context is a physical condition, such as “acrylic mat, weighted puck”. Under context c , the true next state differs from the frozen prediction by an average amount $\mu_c(z, a)$, which we call the *residual*. The training condition has zero residual, $\mu_0 \equiv 0$. Before the first contact, at time t_c , the robot sees the scene and extracts a visual *cue* y_e . Before contact, the robot’s own motion reveals nothing about the context.

Definition 1 (Single-contact task). *A task is single-contact if its success, or a violation of its safety constraint, is decided by the prediction the model makes at the first contact t_c and the action chosen from it. Examples are striking a puck toward a target, tossing an object [93], other fast non-grasping motions [58], lifting an unknown load without jerking or slipping, and pushing an object near a table edge on an unknown surface.*

Observation 1 (Error-driven adaptation cannot help at first contact). *Suppose a method changes its model only in response to prediction errors. Since there are no errors before contact, its prediction at t_c is the same as at the start of the episode. Methods that reset every episode (AdaJEPA, Sandwich-Residuals, gradient updates with reset) therefore make the frozen model’s prediction at first contact. Their error there is $\|\mu_{c_e^*}\|$. On single-contact tasks they cannot do better than the unadapted model.*

The observation follows directly from the definitions. Its purpose is to make clear that on these tasks only two sources of information remain: what the robot has seen in past episodes, and what it sees in the scene before contact.

4 Method: Context-Indexed Residual Adapters

Overview. CIRA has four parts:

- a small set of directions in the latent space where corrections are made;
- one correction model (memory) per known context;
- a probability distribution over which memory is active, updated at every step;
- a learned link between visual cues and memories.

The world model itself is never changed.

Where corrections are made. We collect the prediction errors $r_t = z_{t+1} - f(z_t, a_t)$ of the frozen model on training-condition data. We keep the k directions along which these errors vary most (principal components, with k between 4 and 16), stored as a matrix $U \in \mathbb{R}^{d \times k}$. All corrections are made in these k directions, using the projected error $e_t = U^\top r_t$. The reasoning is that a few physical parameters, such as friction and mass, change the dynamics along only a few directions [19, 43].

Memories. Each context c has a memory: a linear model that predicts the projected error from a small feature vector $\phi_t = \phi(U^\top z_t, a_t)$, for example $\phi_t = [a_t, U^\top z_t, 1]$. The weights W_c are uncertain, and we keep a Gaussian distribution over them:

$$e_t | c \sim \mathcal{N}(W_c \phi_t, \Sigma(z_t, a_t)), \quad W_c \sim \mathcal{MN}(M_c, \Sigma_{\text{row}}, V_c). \quad (1)$$

Here $\mathcal{MN}$ is a matrix normal distribution. It is the matrix version of a Gaussian, so the memory can be updated in closed form after every step [31]. The model includes the action, because a change in friction or motor strength produces an error that grows with the action. A constant offset could not describe this. The correction is also linear in the action, which the safety filter in Section 5 uses. The weights W_c are allowed to drift slowly over time, so each memory can keep learning while it is in use. The memory for the training condition is fixed at $W_0 = 0$. One extra “new context” slot, written $\star$, starts from a broad prior and is used when no stored memory fits.

Noise that depends on the state. The noise covariance $\Sigma(z, a)$ comes from a small network trained on training-condition errors [42]. Around contact the frozen model is already less accurate, for example at the moment of impact or during stick-slip motion, and Σ is larger there. Large errors at those moments are then treated as expected noise rather than as evidence of a new condition. Pushing is known to have this kind of state-dependent noise and variable friction [7, 59]. We found no prior work that links contact noise to the false creation of new contexts. We expect it to be the most common failure of a simple implementation, and our simulated tasks include it on purpose.

Combining factors. A context can consist of several independent factors, such as the surface and the load: $c = (c^1, \dots, c^F)$. Each factor has its own set of memories $W_{c^f}^f$, and the corrections add up: $e_t | c \sim \mathcal{N}(\sum_f W_{c^f}^f \phi_t, \Sigma)$. Two things let the robot separate the factors without labels:

- *Each factor has its own image region.* The surface cue comes from the table area around the object. The load cue comes from a crop of the object or container.
- *Factors change at different times.* People swap mats and containers independently, so the robot sees one factor change while the other stays the same.

Adding the corrections is an approximation. Friction force grows with the normal force, so mass and friction interact. We therefore also keep a memory for each combination the robot has actually seen. For a combination it has not seen, it starts from the sum of the factor memories and refines this memory as data arrive. Our claim about new combinations concerns the prediction at their first contact.

Estimating the active context. At every step the robot updates a probability distribution π_t over the stored memories, the training condition and the “new context” slot. Three sources of evidence are combined:

- *How often each context has occurred.* Before an episode, a context that has been seen often is considered more likely. This is the Chinese restaurant process prior [68]. It gives each known context a probability proportional to how often it has occurred, and gives a new context a small fixed probability.

- *The visual cue*, through the cue model described below.
- *How well each memory predicts the latest step*, through Eq. (1).

Within an episode, contexts are assumed to persist, with a small chance of switching [22]. The update follows the interacting multiple-model filter [12]:

$$\pi_t(c) \propto p(e_t | c, \phi_t) \prod_f p(y^f | c^f) \sum_{c'} p(c | c') \pi_{t-1}(c').$$

This one distribution answers four questions, so no separate change detector is needed:

- if the “new context” slot stays most likely for several steps, a new memory is created;
- if a stored memory is most likely, it is recalled;
- if the training condition is most likely, the robot has returned to normal;
- if the distribution shifts, the condition has changed.

Each memory is updated in proportion to its probability, as in COIN [35], rather than only the most likely one. Two memories that become nearly identical are merged.

Learning what cues mean. For each factor value, the robot keeps a Gaussian model $p(y^f | c^f)$ of what the cue looks like under that context. After each episode, this model is updated using context probabilities computed *from the dynamics only*, without the cue. If the cue model were instead trained on decisions that the cue itself had influenced, a wrong cue could reinforce itself. With this rule, a misleading cue is gradually unlearned.

Optional: priors from vision-language models. A vision-language model that estimates physical properties [25, 84] could set the starting guess for a new context from the image. This would help even the first time a condition is seen. We treat this as an extension and keep it separate from the main claims.

5 Planning and safety in latent space

Planning. We use a sampling-based planner, the cross-entropy method (CEM) or MPPI [17, 72]. Each sampled action sequence is simulated with the frozen model plus a correction, $\hat{z}_{t+1} = f(z_t, a_t) + UW\phi_t$. The weights W come from a context drawn from π_t . The planner can minimise the average cost over contexts, or a risk-averse version (CVaR) that focuses on the worst outcomes [49, 91]. Updating π_t and the memories costs much less than one call of the world model.

Which contexts the safety filter considers. The planner uses the full distribution π_t . The safety filter uses only the contexts with probability at least δ , the *plausible set* $\mathcal{P}_\delta = \{c : \pi_t(c) \geq \delta\}$. This set includes the “new context” slot whenever that slot’s probability exceeds δ . When the robot recognises a known condition, $\mathcal{P}_\delta$ contains a single memory. When it is unsure, or the condition looks new, $\mathcal{P}_\delta$ contains several, and the filter becomes more cautious. As a baseline we also consider a filter that guards against *all* stored memories at once, as in multi-stage model predictive control [56]. We expect that baseline to be much more cautious.

The safety condition. We assume a latent safety function $h(z)$ is given, for example a latent CBF or reachability value trained with existing methods [65, 66]. States with $h(z) \geq 0$ are safe. The filter accepts an action a only if the discrete-time CBF condition [3] holds for every plausible memory:

$$h(f(z_t, a) + UW_c\phi(z_t, a)) - L_h q_t^c \geq (1 - \alpha) h(z_t) \quad \text{for all } c \in \mathcal{P}_\delta. \quad (2)$$

In words, h must not drop by more than a fraction α in one step under any plausible memory, with an extra safety margin $L_h q_t^c$. Here L_h bounds how fast h can change, and q_t^c is an estimate of how wrong memory c can be, explained below. Because the correction is linear in the action, linearising f and h around a planned action turns the condition into one linear constraint per plausible memory. The filter is then a small quadratic program. Over the planning horizon we apply the same condition by discarding sampled action sequences that violate it. If no action satisfies it, the robot stops or retracts.

Safety margins stored with each memory. After each step we measure how far the actual latent state was from the nearest plausible prediction: $s_t = \min_{c \in \mathcal{P}_\delta} \|r_t - UW_c \phi_t\|$. This distance includes ordinary noise, errors outside the k correction directions and mistakes in identifying the context. The margin q_t^c is set by adaptive conformal inference (ACI) [28]. ACI raises the margin after each step where the distance exceeded it and lowers it slightly otherwise, so that in the long run the margin is exceeded on a chosen fraction δ_s of steps. We run a separate ACI for each memory and update it only on steps attributed to that memory. When a known condition returns, its margin is therefore already tuned and does not have to be learned again.

What happens when the cue is wrong. The cue affects π_t and therefore $\mathcal{P}_\delta$. A reliable cue shrinks the plausible set before contact, so the robot can act less cautiously from the start. A misleading cue can leave the true context out of $\mathcal{P}_\delta$. In that case the measured distances grow and ACI raises the margin. The long-run bound in Proposition 3 still holds. A wrong cue therefore costs extra caution and a limited number of extra violations. We measure this cost against a filter that ignores cues.

Optional: safe probing. When the robot is unsure which context is active, it can prefer actions that help tell contexts apart. Among the candidate actions that satisfy Eq. (2) and whose task cost is within a factor $(1 + \epsilon)$ of the best, CEM favours those with the highest expected information about the context, $I(c; e_{t+1:t+H} \mid \mathbf{a})$. This idea comes from active fault diagnosis [10, 11, 75] and dual control [37, 63], combined with safe exploration [9, 44]. In a single-contact task this means the robot gives the puck a small nudge first, but only when the cue and its past experience leave the context unclear. How often it chooses to probe is itself a result we report.

6 Theory

Proposition 1 (Error at first contact). *Consider N episodes whose contexts come from a finite set of K contexts, visual cues that take values in a finite set $\mathcal{Y}$, and memories whose mean predictions are accurate to within ϵ . Let $\Delta_{\max} = \max_{c,c'} \|\mu_c - \mu_{c'}\|$ be the largest difference between two contexts, and $\Delta_{\min} = \min_{c \neq 0} \|\mu_c\|$ the smallest difference between any context and the training condition.*

- (i) *Any method that resets every episode has total first-contact error $\sum_e \|\mu_{c_e^*}\| \geq N_{\text{shift}} \Delta_{\min}$. Here N_{shift} is the number of episodes whose context differs from training.*
- (ii) *In episode e , CIRA has first-contact error at most $\epsilon + (1 - \pi_0^e(c_e^*)) \Delta_{\max}$, where $\pi_0^e(c_e^*)$ is the probability it assigns to the true context before contact. Summed over episodes, $\sum_e (1 - \pi_0^e(c_e^*)) \leq \sum_e -\log \pi_0^e(c_e^*)$. The right-hand side is the total log-loss of predicting each episode’s context from its cue. For any sequence of contexts it is at most $N \hat{H}(C \mid Y) + O(K|\mathcal{Y}| \log N)$ [45], where $\hat{H}(C \mid Y)$ is the empirical conditional entropy of the context given the cue.*

CIRA therefore has lower total first-contact error than any reset-based method whenever $\epsilon + \Delta_{\max} \hat{H}(C \mid Y) < \Delta_{\min} N_{\text{shift}}/N$ (entropy in nats). This ignores lower-order terms and the $O(K)$ episodes in which memories are first learned. It holds for any order of contexts, including adversarial ones.

Proof sketch. Part (i) follows from Observation 1. For (ii), the error of the probability-weighted prediction is $\|\sum_{c \neq c^*} \pi(c)(\mu_c - \mu_{c^*})\| \leq (1 - \pi(c^*)) \Delta_{\max}$, and $1 - x \leq -\log x$ for $x \in (0, 1]$. The log-loss bound is the standard bound for sequential prediction with a Dirichlet mixture, applied separately to each cue value. Creating new contexts adds $O(K \log N)$.

Interpretation. The conditional entropy $\hat{H}(C \mid Y)$ measures how hard it is to guess the context of an episode from its cue and from past frequencies. If conditions repeat and the cue identifies them, it is close to zero and CIRA makes almost no first-contact error. If every episode brings a new condition, memory cannot help and the bound says so. The benefit of the cue is the reduction $\hat{H}(C) - \hat{H}(C \mid Y)$, which is the information the cue carries about the context. The same quantity therefore explains three effects: savings, correct action before contact, and the lack of benefit from an uninformative cue.

Proposition 2 (Steps needed to recognise a known context). *Consider an episode whose true context c^* is stored in memory, and let c' be the most likely alternative. Suppose that after contact each step adds, on average, $D > 0$ to the log-odds in favour of c^* , where D is the KL divergence between the two memories' predictions, and that the step contributions are independent. Before contact they are zero. The starting log-odds combine past frequency and cue: $\lambda_0 = \log \frac{n_{c^*}}{n_{c'}} + \log \frac{p(y|c^*)}{p(y|c')}$, where n_c counts past occurrences of c . If λ_0 already exceeds the decision threshold λ^* , the number of steps needed is zero. Otherwise, by Wald's identity, the expected number of steps is at least $(\lambda^* - \lambda_0)/D$, with equality apart from the final overshoot.*

This gives predictions we can write down before running experiments:

- **Acting before contact.** A learned cue that is correct with probability ρ lets the robot correct about a fraction ρ of the error before contact.
- **Savings.** Each further occurrence of a context shortens recognition by $\log(n_{k+1}/n_k)/D$ steps, so the gains shrink over time. For gradient updates with reset, the time stays the same.
- **Misleading cues.** A misleading cue of strength ℓ_y (its log-likelihood ratio) costs about $2\ell_y/D$ extra steps compared with a correct cue.
- **Noise.** The state-dependent noise model lowers D around contact. This makes recognition slower but prevents false new contexts. We measure both effects. For the case without cues, classical results on detection delay [46, 55] serve as a reference.

Proposition 3 (Long-run safety for any cue quality). *Suppose Eq. (2) is enforced exactly whenever some action satisfies it (for example by rejecting sampled actions), and the fallback action is safe otherwise. Suppose ACI runs separately for each memory with target rate δ_s and step size γ . Then, for any sequence of contexts and cues, the one-step condition $h(z_{t+1}) \geq (1 - \alpha)h(z_t)$ fails on at most a fraction $\delta_s + O(K/(\gamma T))$ of the first T steps.*

Proof sketch. If the measured distance s_t is within the margin q_t^c , the true next state lies within q_t^c of the prediction of some plausible memory. Eq. (2) and the bound L_h on how fast h changes then give the one-step condition. So the condition can fail only on steps where the margin was exceeded. For each memory, ACI guarantees that the long-run fraction of such steps is at most $\delta_s + O(1/(\gamma T_c))$, without assumptions on the data distribution [28]. Summing over the K memories gives the result.

The bound applies to the safety condition in latent space. Whether this implies physical safety depends on whether h correctly represents the real constraint, which we treat as a property of the encoder [57]. In single-contact tasks it is natural to count episodes instead of steps, and the bound becomes a long-run rate of unsafe episodes.

7 Measuring recall versus learning

For every adaptive method in this proposal, we separate its adaptive state into two parts:

- *which memories it uses*, π : for CIRA this is the context distribution;
- *what the memories contain*, Θ : the weights.

Methods that reset every episode have only Θ . Let $L(\pi, \Theta)$ be the prediction error on data from the current condition, computed by replaying recorded transitions.

Definition 2 (Splitting recovery into recall and learning). *After a change at time t_0 , the improvement by time t is $R_t = L(\pi_{t_0}, \Theta_{t_0}) - L(\pi_t, \Theta_t)$. The part due to changing which memories are used is*

$$A_t = \frac{1}{2} [L(\pi_{t_0}, \Theta_{t_0}) - L(\pi_t, \Theta_{t_0})] + \frac{1}{2} [L(\pi_{t_0}, \Theta_t) - L(\pi_t, \Theta_t)],$$

the average effect of switching π with Θ held at its old and at its new value (a Shapley value with two players). We call A_t apparent learning. The remainder $P_t = R_t - A_t$ is proper learning.

The two parts always add up to the total, and the result does not depend on which part is computed first. For CIRA every term can be computed in closed form from stored distributions. For methods that reset every episode, A_t is always zero. We apply the same split to the safety margin. This shows how much of the reduction in caution after a change comes from recalling a memory and how much from recalibration.

8 Hypotheses and stopping criteria

- H1 First contact.** On single-contact tasks with recurring conditions and an informative cue, CIRA succeeds more often than the frozen model and all error-driven baselines. Its first-contact error follows Proposition 1.
- H2 Savings.** Recognition time falls with each further occurrence, as predicted by Proposition 2. For gradient updates with reset it does not change.
- H3 No interference.** Learning condition B does not reduce performance on condition A. Continuously running gradient updates and continual test-time adaptation do reduce it.
- H4 Misleading cues.** The observed dynamics overrule a misleading cue within the predicted number of steps. Task cost stays below that of the frozen model, and the safety bound holds.
- H5 New combinations.** For a combination of factors never seen together (for example the acrylic mat with the weighted puck), the combined factor memories give lower first-contact error than both the version without cues and the version without factors.
- H6 Safety.** The long-run rate of violations of the latent safety condition stays within δ_s plus the predicted slack, also with misleading cues. At the same violation rate, our filter intervenes less than a filter that guards against all memories and less than a filter with one shared margin. Its caution decreases as conditions recur.
- H7 Noise.** With noisy contacts, no single learning rate for gradient updates is best both in normal conditions and after a change. CIRA matches the best rate in each phase. Its rate of falsely created contexts stays below a limit fixed in advance.
- H8 Recall versus learning.** In the long real deployment, recall accounts for a substantial share of recovery. Fixed in advance: the median of A/R is at least 0.3.

When we change the plan.

- If the version without cues is as good as the full method at first contact, we drop the claim about acting before contact. The paper then focuses on memory, the benchmark and the recall/learning measurement.
- If well-tuned gradient updates with reset are as good as the version without cues on recurring conditions, the method offers no advantage. The paper then reports the benchmark and this negative result.
- If combining factors fails on unseen combinations, we report whole-context memories only and drop the claim about new combinations.
- If our safety filter is not less cautious than the all-memories filter at the same violation rate, we present the safety part as a supporting result rather than a main contribution.

9 Experiments

Sequences of conditions. Each condition lasts for a random number of episodes (geometric distribution), and it occasionally changes within an episode. Each condition has a matching appearance. The cue shows that appearance with probability ρ and a different one otherwise. We test $\rho \in \{1/M, 0.5, 0.65, 0.8, 0.95\}$, where M is the number of conditions and $\rho = 1/M$ means the cue carries no information. The sequences contain four patterns taken from motor-learning experiments:

- *savings*: conditions A, B, A, B in blocks;
- *interference*: a long or a short block of A before B;
- *evoked recovery*: two episodes of A inserted into a block of normal conditions;
- *misleading cues*: the pairing between cue and condition swapped for one block.

Each pattern is linked to one of the predictions above. A pattern that does not separate the methods in simulation is removed.

Environments.

- (i) **Simulation.** A 2D pushing task and a 2D puck strike with a table edge. Contacts are noisy, and surface and load vary. Cues are correct 80% of the time. These tasks run on a laptop and decide whether H1, H2, H4, H5 and H6 hold before we invest in hardware.
- (ii) **AdaJEPA tasks** [86]. PushT, PushObj and PointMaze with the released JEPA and DINO-WM [94] models. We pair the existing colour changes of the agent or block with changes in dynamics, so that colour becomes a cue, and we add random noise to the dynamics. This allows a direct comparison with AdaJEPA and Sandwich-Residuals.
- (iii) **Real robot, single-contact tasks.**
 - *Strike*: strike a puck to a target on felt, rubber or acrylic mats, without it sliding off the table edge [92].
 - *Lift and place*: lift loads in visually distinct containers without exceeding a limit on jerk or tilt.

The world model is a frozen V-JEPA 2-AC [6] or a DINO-WM-style model trained on training-condition data. The memories add only a few kilobytes on top of a large pretrained model that stays unchanged.

- (iv) **Real robot, long deployment.** At least 1000 episodes over several days. People swap mats and containers according to a schedule the robot does not know, and sometimes cover the mats with a neutral sheet to make cues less reliable. We report the recall/learning split, predicted versus observed first-contact error, and the violation rate.

Baselines.

- Adaptation:
 - the frozen model;
 - continuously running gradient updates (as in AdaJEPA);
 - gradient updates with periodic reset [70];
 - gradient updates reset exactly at each change (an idealised version);
 - Sandwich-Residuals [78];
 - a MOLe-style mixture of models [64];
 - MOCA-style change detection without recall [32].
- Safety:
 - the uncertainty-aware latent filter [73];
 - a latent filter with one shared ACI margin [40];
 - a filter that guards against all memories [56].
- Ablations of CIRA:
 - without cues;
 - without factors;
 - with constant noise;
 - with a constant-offset memory;
 - updating only the most likely memory;
 - with one shared margin instead of one per memory.
- Upper bound: CIRA told the true context.

All settings are tuned on training-condition data only. Random seeds and sequences are fixed and published before the main experiments.

Metrics.

- Error at first contact, and success on single-contact tasks.
- Number of steps after contact needed to recognise the condition.
- Total extra prediction error after each change.
- Savings: recognition time for the k -th occurrence relative to the first.
- Performance on condition A after learning condition B.
- Error on unseen combinations of factors.
- Violation rate, and how much the safety filter changes the planned action or raises task cost.
- The recall share A/R , for prediction error and for the safety margin.
- Rates of falsely created and missed contexts.
- Predicted versus observed values for Propositions 1 and 2.

10 Risks

- **Contact noise creates false contexts.** This is the most likely problem. The state-dependent noise model and a minimum number of steps before a new memory is created address it. We report the rate of false contexts as a main metric.
- **Cues are weak.** We vary cue reliability. Proposition 1 predicts how much a weak cue should still help, so weak cues yield a result rather than a failure.
- **Factors interact.** Mass and friction are expected to interact. Keeping memories for seen combinations limits the damage, and the test on unseen combinations measures how large the problem is.
- **A change falls outside the correction directions.** The safety margin is measured in the full latent space, so safety does not depend on the chosen directions. For prediction, we monitor the error outside these directions and add directions for a new memory when this error grows.
- **Latent safety differs from physical safety.** Proposition 3 concerns the latent condition. We report physical violations separately. Whether h matches the real constraint is outside the scope of this proposal [57].
- **Competing work.** AdaJEPA and Sandwich-Residuals name persistence as future work, and the latent ACI filters [40, 67] appeared this year. Our distinguishing points are acting before contact, combining factors, the first-contact analysis and margins stored per memory. A workshop paper by February 2027 establishes priority.
- **The motor-learning patterns add nothing.** Each pattern is tied to a numerical prediction and is removed if it does not separate the methods.

11 Timeline

When	Work	Done when
Oct 2026	Simulated pushing and puck strike; core of CIRA; gradient baselines	H1, H2 and H4 visible; stopping criteria checked
Nov 2026	Factor memories; safety filter with per-memory margins; recall/learning measurement	H5 and H6 in simulation; Propositions 1–3 compared with simulation
Dec 2026–Jan 2027	AdaJEPA tasks with all baselines; real-robot setup (mats, containers, puck, edge)	Condition sequences running in PushT and PointMaze
Feb 2027	Workshop paper: simulation, AdaJEPA tasks, theory	Submitted
Mar–Apr 2027	Real-robot single-contact tasks; long deployment (≥ 1000 episodes); safe probing	H1–H8 tested on hardware
Spring 2027	Full paper for CoRL 2027 (deadline to be confirmed when announced)	Submitted

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